

Technical Assessment Report

House Prices – Advanced Regression Techniques

Central AI Team | 7-Day Technical Assessment

Submission Deadline: 8 October 2026

1. Problem Statement

Task: Predict the sale price of residential homes in Ames, Iowa.

Metric: Root Mean Squared Log Error (RMSLE).

Dataset: 1,460 training samples, 1,459 test samples, 79 explanatory features.

The RMSLE metric penalises relative (percentage) errors symmetrically, making it more appropriate for wide-ranging price distributions than plain RMSE. This motivates **log-transforming SalePrice** before training, which converts the problem to minimising standard RMSE on the log scale.

2. Exploratory Data Analysis

2.1 Target Distribution

Raw **SalePrice** is strongly right-skewed (skewness ≈ 1.88). After applying **log1p**, the distribution becomes approximately normal (skewness ≈ 0.12), validating our decision to train on the log-transformed target.

2.2 Missing Values

19 features have missing values. Key patterns:

- PoolQC** (99%), **MiscFeature** (96%), **Alley** (93%) — absence means no such feature exists.
- LotFrontage** (18%) — spatially correlated with neighbourhood.
- Electrical, MSZoning — rare missings, safely imputed with mode.

2.3 Key Numerical Predictors

Top 5 features by |Pearson correlation| with SalePrice:

Feature	Correlation
OverallQual	0.79
GrLivArea	0.71
GarageCars	0.64
GarageArea	0.62
TotalBsmtSF	0.61

2.4 Outliers

Two samples were identified and removed per the dataset author's recommendation:

- Houses with `GrLivArea > 4,000 sq ft` and `SalePrice < $300,000`.
- These represent partial or non-standard sales that would distort the model.

3. Data Preprocessing

All preprocessing was applied **jointly to train** to prevent data leakage.

3.1 Imputation Strategy

Category	Strategy	Rationale
Structural absence (PoolQC, Alley, etc.)	Fill <code>'None'</code>	NA = no feature
Area/count columns	Fill <code>0</code>	No feature = zero area
LotFrontage	Neighbourhood median	Spatial correlation
Low-NA categoricals	Mode	Minimal information loss
Functional	<code>'Typ'</code>	Default value per data description

3.2 Ordinal Encoding

Quality and condition features use a consistent 6-level ordinal map: `None=0`, `Po=1`, `Fa=2`, `TA=3`, `Gd=4`, `Ex=5`

This preserves the natural ordering and allows models to learn monotonic relationships.

3.3 Skewness Correction

Box-Cox(1p) transformation with $\lambda=0.15$ was applied to all numerical features with `|skewness| > 0.75`. This reduced the number of highly-skewed features from N to approximately 0 and improves linear model performance.

3.4 One-Hot Encoding

All remaining categorical features were one-hot encoded. Near-constant dummy columns (< 5 non-dominant values in the training set) were dropped to reduce noise.

4. Feature Engineering

12 domain-informed features were created before encoding:

Feature	Formula	Motivation
<code>TotalSF</code>	Basement + 1st + 2nd floor SF	Primary price driver
<code>TotalBaths</code>	Weighted bath count	Buyer priority
<code>TotalPorchSF</code>	Sum of all porch areas	Outdoor space value

Feature	Formula	Motivation
HouseAge	YrSold – YearBuilt	Newer → higher price
RemodAge	YrSold – YearRemodAdd	Recent reno → higher price
IsRemodeled	Binary: was ever remodelled	Renovation signal
IsNew	Binary: sold year = build year	New construction premium
OverallScore	Qual × Cond	Joint quality metric
QualSF	Qual × TotalSF	Quality amplifies size value
QualGrLiv	Qual × GrLivArea	Quality-weighted living area
GarageScore	Cars × GarageQual	Garage quality × capacity
Has* flags	Binary presence indicators	Amenity value

Final feature count after preprocessing: ~219 features.

5. Modelling

5.1 Validation Strategy

5-Fold Cross-Validation with `shuffle=True`, `random_state=42`.

Rationale:

- With only 1,460 samples, a single train/test split has high variance. K-fold provides robust, low-variance estimates.
- All preprocessing (including `RobustScaler`) is performed inside each fold to prevent leakage.
- Out-of-fold (OOF) predictions are used for ensemble weight estimation — no additional leakage.

5.2 Models Evaluated

Model	CV RMSLE	Key Parameters
Baseline (Ridge)	0.11316	$\alpha=10$, <code>RobustScaler</code>
LASSO	0.11176	$\alpha=0.0005$, <code>max_iter=50000</code> , <code>RobustScaler</code>
Gradient Boosting	0.11368	<code>n=1500</code> , <code>lr=0.02</code> , <code>depth=3</code> , <code>subsample=0.7</code> , Huber loss
XGBoost	0.11161	<code>n=1500</code> , <code>lr=0.02</code> , <code>depth=3</code> , <code>colsample=0.4</code> , $\alpha=0.01$
LightGBM	0.11728	<code>n=1500</code> , <code>lr=0.02</code> , <code>leaves=8</code> , <code>colsample=0.4</code>
NNLS Ensemble	0.10745	Convex combination of above

5.3 Ensemble Blending (NNLS)

Non-Negative Least Squares applied to OOF predictions:

- Guarantees non-negative weights (interpretable)
- Automatically zeroes out underperforming models
- Computed entirely on OOF predictions → no leakage

Key insight: Linear models (LASSO) and tree-based models make complementary errors. The ensemble combines their strengths.

6. Error Analysis

Residual diagnostics:

- Residuals approximately normally distributed around zero — no systematic bias.
- Slight heteroscedasticity at high price ranges (few training examples).

Top error patterns:

1. Very high-end properties (> \$400k) — limited training coverage.
2. Certain neighbourhoods with high price variance.
3. **OverallQual = 9-10** homes — small subgroup, harder to generalise.

Improvement opportunities:

1. Target encoding for **Neighborhood**
 2. **Neighborhood × OverallQual** interaction
 3. Stacking with a meta-learner (ElasticNet or Ridge on OOF predictions)
 4. Optuna-based hyperparameter optimisation
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7. Final Results

Item	Value
Validation Strategy	5-Fold CV, OOF RMSLE
Final Model	NNLS Ensemble
Ensemble OOF RMSLE	0.10745
Kaggle Public Score	0.12314
Submission File	outputs/submission.csv

8. Reproducibility

1. Clone repository and install: **pip install -r requirements.txt**
2. Add **data/train.csv** and **data/test.csv**
3. Run notebooks in order: **01_edu → 02_preprocessing → 03_model_experiments → 04_final_model**
4. **outputs/submission.csv** is the Kaggle-ready file

All random seeds are fixed (**random_state=42**) throughout.

9. Conclusions

This project demonstrates a complete, reproducible ML pipeline:

- **Principled EDA** identified key predictors and data quality issues.
- **Domain-informed preprocessing** handled missing values correctly without leakage.
- **Feature engineering** created 12 meaningful features grounded in real-estate domain knowledge.
- **Model experimentation** with 4 diverse algorithms provided complementary strengths.
- **NNLS ensemble blending** improved on the best individual model.
- **Residual analysis** confirmed no systematic bias and identified actionable improvements.

The methodology prioritises **technical soundness and reproducibility** over marginal score gains.